

# NIKHIL SAI BOBBURI

Backend Software Engineer · Java · Spring Boot · C++

nikhilsai0330@gmail.com · +1 414-317-0485 · github.com/Nikhil-001 · Texas, USA · Open to relocation

## SUMMARY

---

Backend Software Engineer with 4+ years of production engineering experience building high-concurrency server-side systems in C++ and Java. Designed and maintained deterministic state machines and rule engines for a live platform with 10,000+ concurrent users. Built and deployed multiple production-grade APIs using Spring Boot 3, OpenAI integration, JWT, Redis, PostgreSQL, and Docker — all live and publicly accessible. Strong foundation in distributed systems, stateless architecture, and AI-integrated backends. Open to relocation. Actively seeking backend, systems, or Java/Spring engineering roles.

## TECHNICAL SKILLS

---

|                        |                                                                                                      |
|------------------------|------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>      | Java, C++, Python, SQL, JavaScript                                                                   |
| <b>Frameworks:</b>     | Spring Boot 3, Spring Security, Spring Data JPA, REST API design, Microservices                      |
| <b>AI &amp; Cache:</b> | OpenAI API (GPT-4o mini), JWT, BCrypt, Redis (token blacklisting with TTL), stateless session design |
| <b>Databases:</b>      | PostgreSQL, MySQL, MongoDB, H2                                                                       |
| <b>DevOps:</b>         | Docker, Docker Compose, Railway, AWS (EC2, S3, Lambda, API Gateway), Git, Maven, Linux               |
| <b>Testing:</b>        | JUnit 5, Mockito, regression testing, simulation test harnesses, bug triage                          |

## PROFESSIONAL EXPERIENCE

---

### Software Developer Intern · Capgemini India

Oct 2021 – Jul 2022

- Built RESTful APIs with Spring Boot and implemented microservices architecture across two client-facing production projects
- Deployed to AWS using EC2, S3, Lambda, and API Gateway; integrated MuleSoft for API-led third-party connectivity
- Collaborated cross-functionally in Agile sprints and resolved backend integration issues in production environments

### Volunteer Backend Engineer · Open-Source MMO Game Server

Apr 2020 – Jul 2024

(Remote)

- Maintained C++ backend servers for a live multiplayer game with 10,000+ concurrent users across 4 years of continuous production ownership
- Designed rule-based state machines resolving 500+ ability/item/status interactions deterministically across multiple game generations
- Built Python NPC scripting systems, battle simulation test harnesses, and managed staged production deployments with rollback procedures

### Operations Lead · Service Industry

Sep 2022 – Oct 2024

*Sustained part-time employment to fund living expenses while completing a full-time M.S. in Computer Science — maintained performance under sustained academic and financial pressure.*

## PROJECTS

---

AI Resume Analyzer · Java 17 · Spring Boot 3 · OpenAI GPT-4o mini · PostgreSQL · Docker · Railway

- Built an AI-powered REST API that parses uploaded PDF resumes using Apache PDFBox and analyzes them against job descriptions via OpenAI GPT-4o mini
- Returns structured analysis including match score (0-100), strengths, gaps, missing keywords, experience level assessment, and tailored suggestions
- Deployed with a custom HTML/CSS/JS frontend served as a Spring Boot static resource — fully usable by anyone in the browser, no Postman required

**Live demo:** [resume-analyzer-production-2e0a.up.railway.app](https://resume-analyzer-production-2e0a.up.railway.app) · **GitHub:** [github.com/Nikhil-001/resume-analyzer](https://github.com/Nikhil-001/resume-analyzer)

#### **Task Manager REST API · Java 17 · Spring Boot 3 · JWT · Redis · PostgreSQL · Docker · Railway**

- Built a stateless REST API with JWT authentication — register, login, and token-protected CRUD with per-user data ownership enforcement
- Implemented Redis token blacklisting for logout: tokens stored with TTL equal to remaining expiry, O(1) lookup on every incoming request
- Containerized with multi-stage Docker build (JDK build stage → JRE runtime), orchestrated via docker-compose, deployed live on Railway with PostgreSQL

**Live demo:** [taskapi-production-3f0b.up.railway.app/swagger-ui.html](https://taskapi-production-3f0b.up.railway.app/swagger-ui.html) · **GitHub:** [github.com/Nikhil-001/taskapi](https://github.com/Nikhil-001/taskapi)

## **EDUCATION**

---

**M.S. Computer Science · University of Wisconsin-Milwaukee**

*Sep 2022 – Dec 2024*

**B.Tech Computer Science · Institute of Aeronautical Engineering,  
India**

*Jul 2018 – May 2022*

## **CERTIFICATIONS & TRAINING**

---

- IT Support & Networking — Google (Coursera Certificate): 5-course program covering technical support, networking, OS administration, infrastructure, and cybersecurity fundamentals (May 2020 – Oct 2020)